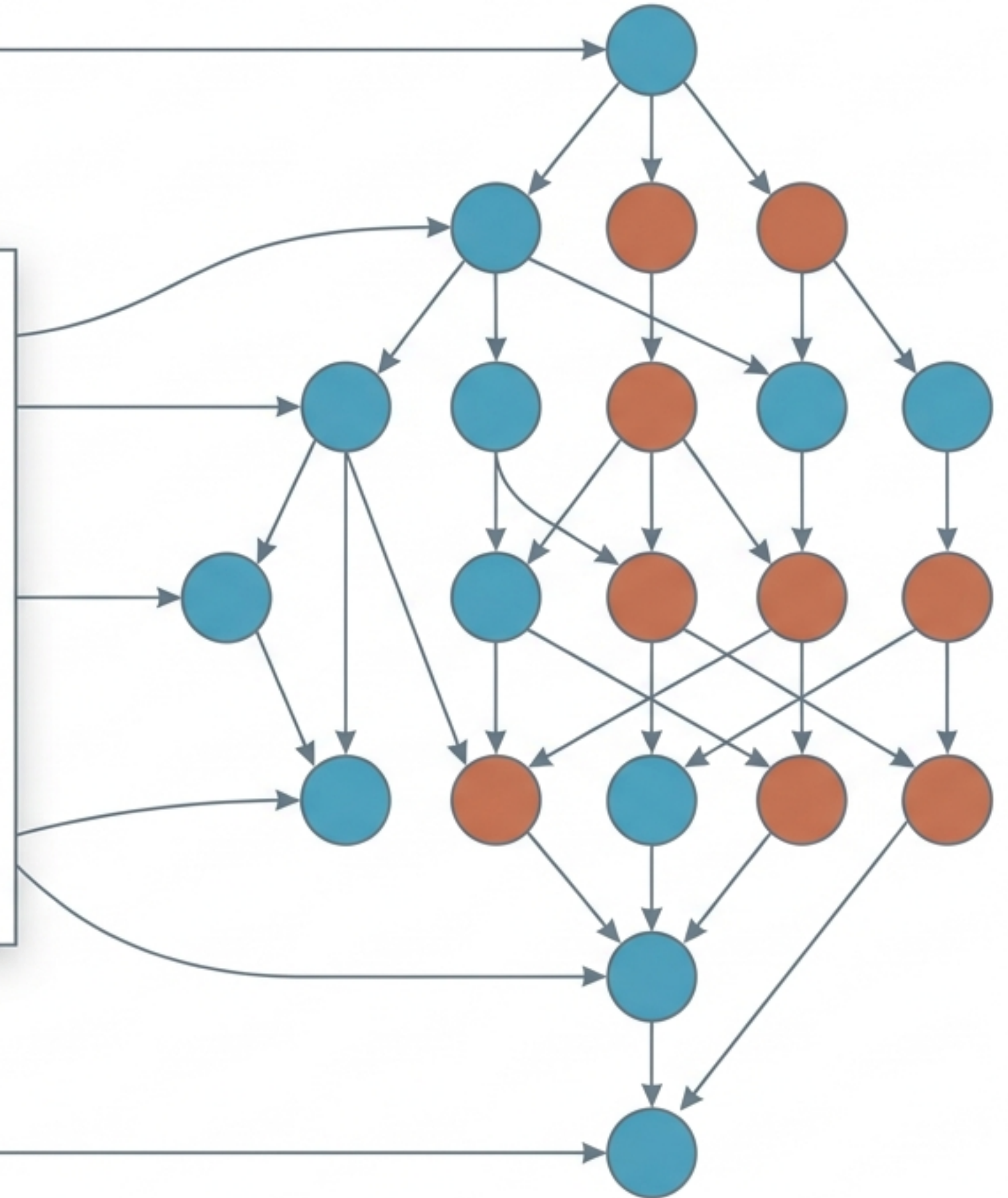


Tying the Data Knot

A State-of-the-Art Machine Learning Pipeline for Digital Romance

OCC6 Group 3 | University of Malaya | V8 Patched Edition



Chew Wei Jian

Pipeline Parallelization,
Integration & Thread Caching

Ng Jin Ru

Exploratory Data Analysis &
Class Balancing

Chaang Wai Chiu

SHAP Explainability, UI
Dashboard & Fairness Audits

GitHub Central Repository

**Merge
(V8.0)**

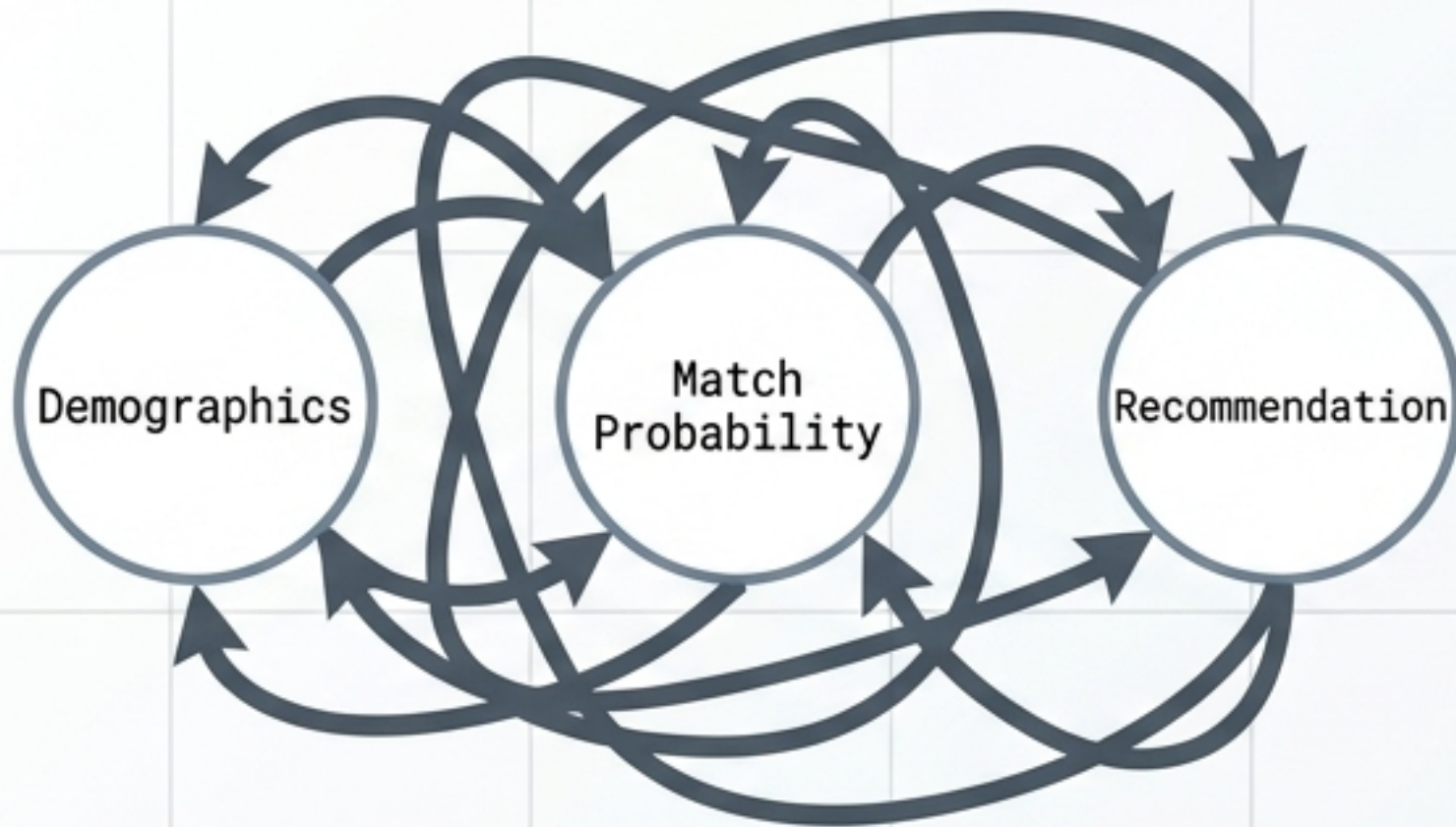
Ku Jian Cheng

Data Preprocessing &
Categorical Encoding

Ang Ying En

Model Optimization &
RandomizedSearchCV Grids

Traditional ML Matchmaking



- **Goal:** Purely Predictive
- **Flaw:** Learns spurious demographic correlations (e.g., location/income) rather than true compatibility.
- **Result:** High churn, ghosting, and backdoor confounding.

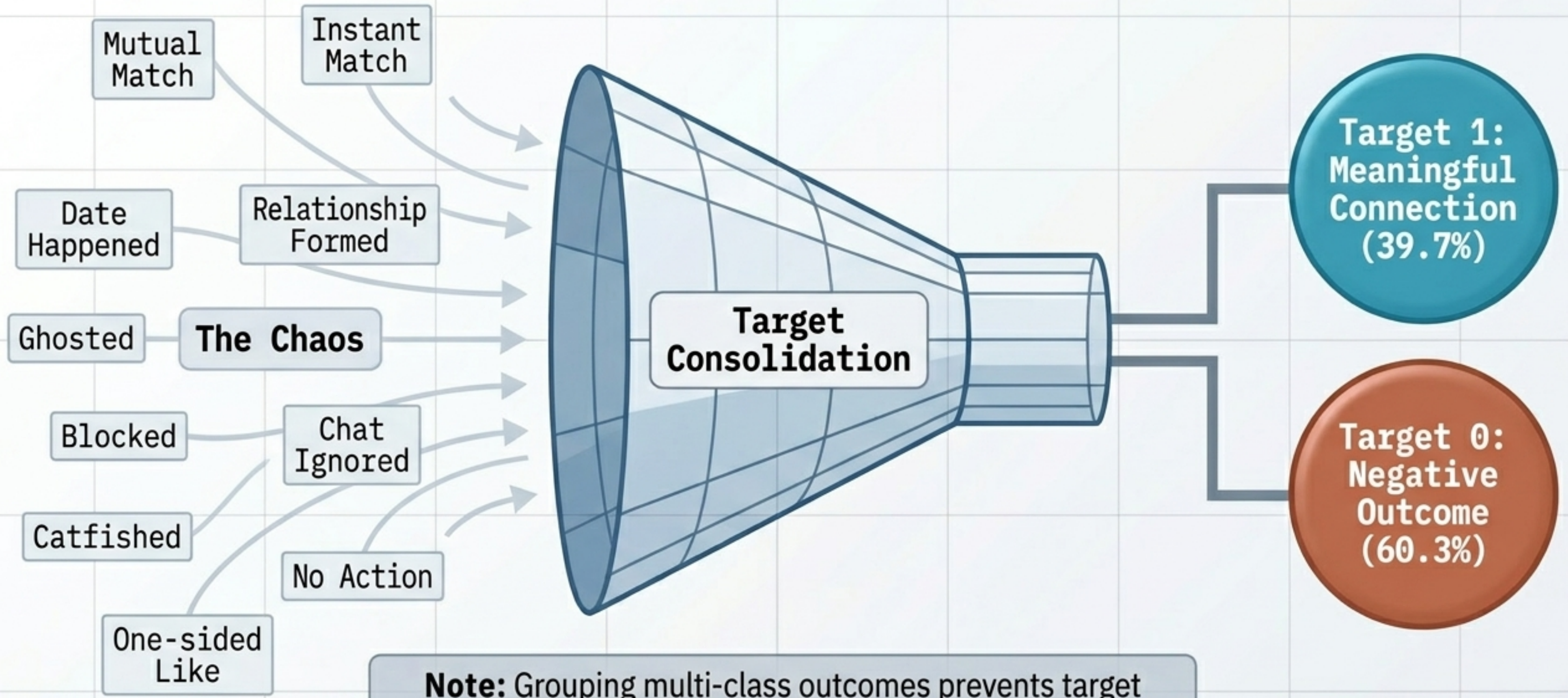
The V8 Patched Edition



- **Goal:** Prescriptive & Counterfactual
- **Infrastructure:** Leakage-Free Causal Audit Sandbox.
- **Result:** Mathematically isolated Average Treatment Effects (ATE) mapping true behavioral signals.

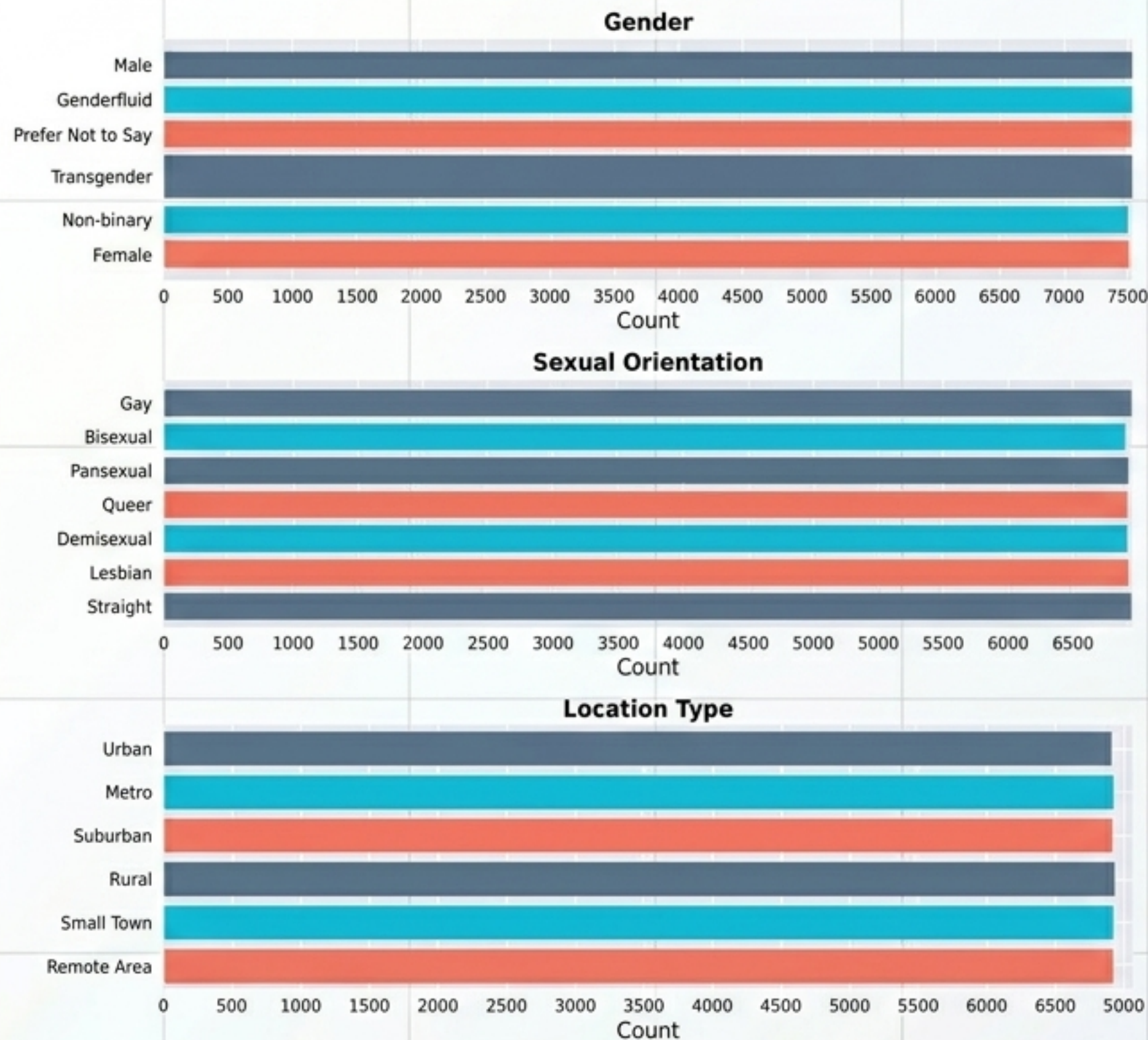
Transitioning from simple prediction to ethical, causal prescription.

50,000 Samples | 25 Feature Dimensions



Note: Grouping multi-class outcomes prevents target sparsity and streamlines causal uplift analysis.

EDA: Categorical Uniformity

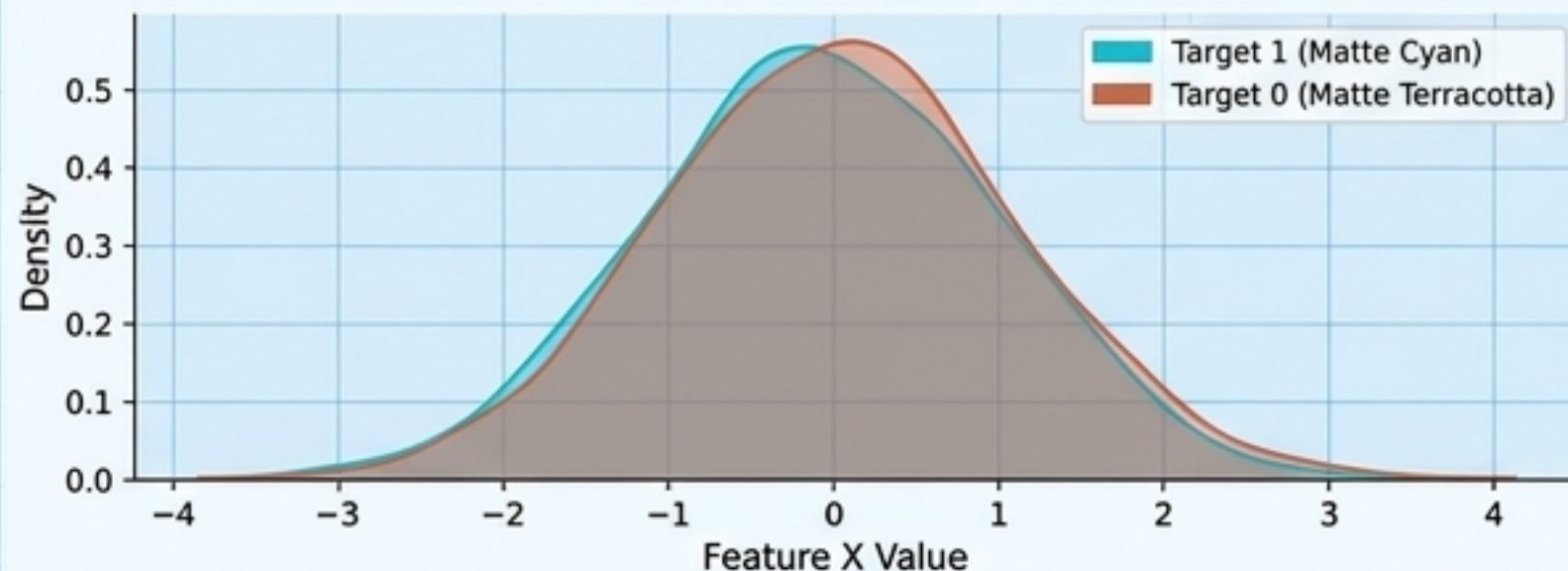


The Mathematical Advantage

- ✓ Perfectly flat categorical distributions prevent demographic dominance in the loss function.
- ✓ Establishes a pristine, unbiased base for strict downstream Demographic Parity fairness audits.
- ✓ Models evaluate all cohorts with equal weight, bypassing standard real-world sample skew.

EDA: The Case for Robust Scaling

Univariate Linear Signal Analysis



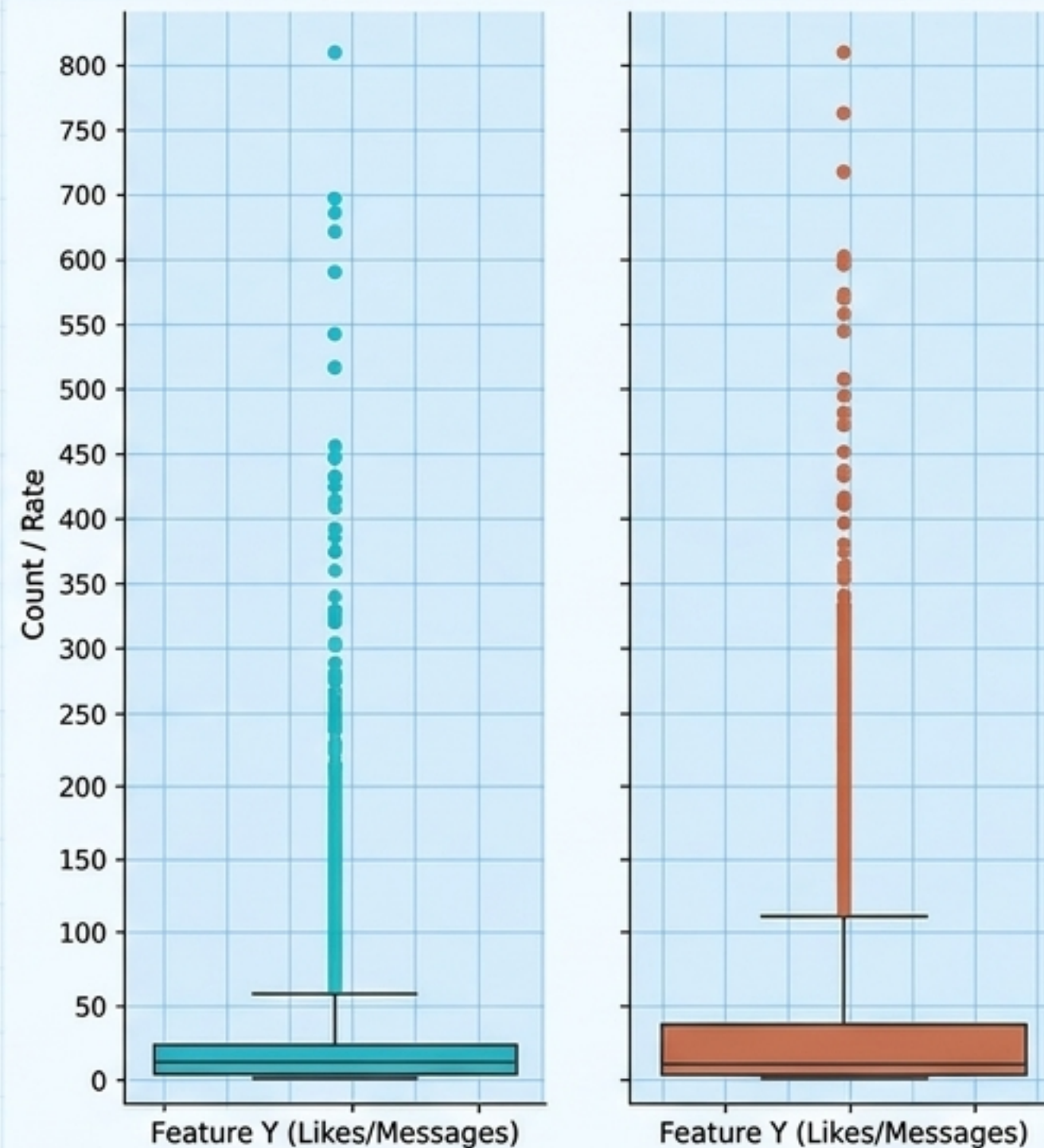
Target 1 & 0: Identical Medians = No Univariate Linear Signal

[PIPELINE UPGRADE: RobustScaler]

Standard mean-variance scaling collapses under heavy-tailed outliers. V8 utilizes median-based scaling.

V8 Patch Note: Fitting strictly deferred until AFTER train-test split to guarantee zero data leakage. ■

Outlier Detection



Extreme Power-User Outlier Noise